

Ex. 5: MIMO System Simulation with BER Analysis

Code

```
clc;
clear;
close all;
numBits = 1e5;
SNRdB = 0:2:20;
bps = 2;
Nt = input('Transmit antennas Nt = ');
Nr = input('Receive antennas Nr = ');
bits = randi([0 1],numBits,1);
bits2 = reshape(bits,2,[]).';
symbols = (2*bits2(:,1)-1) + 1i*(2*bits2(:,2)-1);
symbols = symbols/sqrt(2);
extra = mod(length(symbols),Nt);
if extra~=0
symbols = symbols(1:end-extra);
bits = bits(1:2*length(symbols));
end
numSym = length(symbols);
sym_tx = reshape(symbols,Nt,[]).';
BER = zeros(size(SNRdB));
for s = 1:length(SNRdB)
snr = 10^(SNRdB(s)/10);
noiseVar = 1/snr;
numBlocks = size(sym_tx,1);
errors = 0;
totalBits = 0;
for k = 1:numBlocks
H = (randn(Nr,Nt)+1i*randn(Nr,Nt))/sqrt(2);
noise = sqrt(noiseVar/2)*(randn(Nr,1)+1i*randn(Nr,1));
y = H*sym_tx(k,:).' + noise;
xhat = pinv(H)*y;
rxBits = zeros(2*Nt,1);
rxBits(1:2:end) = real(xhat)>0;
rxBits(2:2:end) = imag(xhat)>0;
txBits = zeros(2*Nt,1);
for m=1:Nt
txBits(2*m-1)=real(sym_tx(k,m))>0;
txBits(2*m)=imag(sym_tx(k,m))>0;
end
errors = errors + sum(rxBits~=txBits);
totalBits = totalBits + 2*Nt;
end
BER(s)=errors/totalBits;
end
figure;
semilogy(SNRdB,BER,'-o','LineWidth',2);
```

```

grid on;
xlabel('SNR (dB)');
ylabel('Bit Error Rate');
title(sprintf('%d×%d MIMO using QPSK with Zero-Forcing',Nt,Nr));

```

